escape such “compressed stars,” they were initially called dark or frozen stars. A more catchy name was coined years later by John Wheeler, who called them black holes—black because they cannot emit light, holes because anything getting too close falls into them, never to return. The name stuck.

We illustrate Schwarzschild’s solution in Figure 3.7. Although black holes have a reputation for rapacity, objects that pass by them at a “safe” distance are deflected in much the same way that they would be by an ordinary star, and can proceed on their merry way. But objects of any composition whatsoever that get too close—closer than what has been termed the black hole’s event horizon—are doomed: they will be drawn inexorably toward the center of the black hole and subject to an ever increasing and ultimately destructive gravitational strain. For example, if you dropped feet first through the event horizon, as you approached the black hole’s center you would find yourself getting increasingly uncomfortable. The gravitational force of the black hole would increase so dramatically that its pull on your feet would be much stronger than its pull on your head (since in a feet-first fall your feet are always a bit closer than your head to the black hole’s center); so much stronger, in fact, that you would be stretched with a force that would quickly tear your body to shreds.

If, on the contrary, you were more prudent in your wanderings near a black hole and took great care not to trespass beyond the event horizon, you could make use of the black hole for a rather amazing feat. Imagine, for example, that you were to discover a black hole whose mass was about 1,000 times the mass of the sun, and that you were to lower yourself on a cable, much as George did near the sun, to about an inch above the black hole’s event horizon. As we have discussed, gravitational fields cause a warping of time, and this means that your passage through time would slow down. In fact, since black holes have such strong gravitational fields, your passage through time would slow way down. Your watch would tick about ten thousand times more slowly than those of your friends back on earth. If you were to hover just above the black hole’s event horizon in this manner for a year, and then climb up the cable to your waiting starship for a short, yet leisurely, journey home, upon arrival at earth you would find that more than ten thousand years had passed since your initial